should be connected to two-third of the voltage of the maximum voltage.

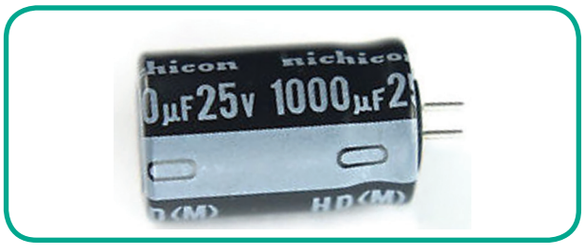


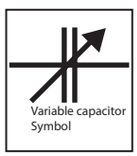

Figure 1.63(a) Working voltage

In the Figure 1.63(a), 1,000 micro farad / 25volts is mentioned in the capacitor. 25 V is the maximum voltage to be applied. Two third of 25 V, that is 16 V should be normally applied to this capacitor.

## 1.11.8 Variable Capacitor

In variable capacitor, the capacitance value can be changed within a particular limit.

In this section, different types of variable capacitors used in electronics circuit are discussed.




### 1.11.8.1 Ganged Capacitor

Two variable capacitors are arranged in a setup and can be varied through a common shaft is called Ganged capacitor.

It is of two types:

1. Metal gang
2. PVC gang

#### 1.11.8.1.1 Metal Gang

Figure 1.64 shows a ganged capacitor.In this, there are two sets of plates and air acts as dielectric medium. One is static (stator) and another set is kept rotating (rotor). Rotor is connected to a shaft. When the shaft is rotated, metal plates mounted on the shaft, moves between fixed metal plates. This varies the value of the capacitance.

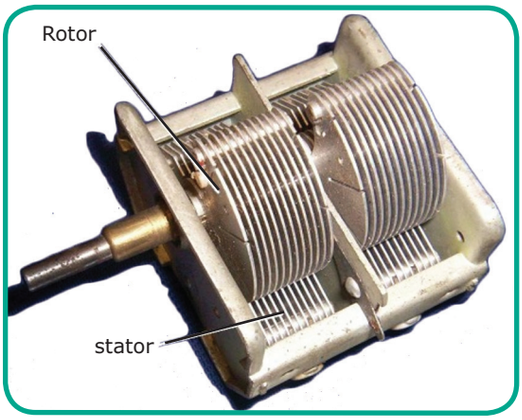



Figure 1.64 Ganged capacitor (Metal)

If rotor moves outward, the area between stator and rotor is less, capacitance also becomes low value. If rotor moves inside the stator, the area becomes large. Hence capacitance value also becomes high. The minimum capacitance is 49 pF and maximum is 500pF. It consists of two variable capacitors as shown in the Figure 1.64.

#### 1.11.8.1.2 PVC Gang

In this type of gang, thin layer of PVC acts as a dielectric medium.

Rotor is connected with chassis. Its value depends upon the area of plates. It is shown in figure 1.65.

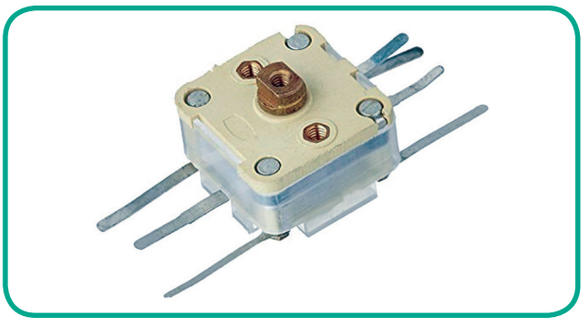


Figure 1.65 PVC gang

### Uses

It is used in tuning circuits of the radio receivers for selecting a desired radio station.



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